

MERCHANT SERVICES · PRODUCT AND TECHNOLOGY LEADERSHIP

Decision Platform

Continuously optimize every payment decision — with evidence.

Move routing, credential, authentication and retry optimization from release-driven change to measurable, reversible strategies.

September 2026 · Executive summary, 15 slides

Methodology, worked example and full requirements are in the companion deep-dive deck.

The constraint is not a shortage of ideas

We have no shortage of proposals for improving approval rates and acceptance cost. What we lack is a reliable way to change one payment decision, expose it safely, measure what it caused, and keep the learning. Three costs follow from that, and all three are structural.

Slow

A routing or credential change is a module release. Weeks to quarters — and the release re-tests everything else in the module, so the cost is not proportional to the change.

Unprovable

We compare before and after. That confounds the change with seasonality, mix shift, issuer behaviour and every other change shipped the same month. We cannot say what anything was worth.

Unrepeatable

No counterfactual is recorded, so there is nothing to train a model on and nothing to evaluate one against. Every model proposal starts from zero evidence.

None of these is a tooling gap. They are all consequences of the unit of change being a module version rather than a decision.

From the module release to the decision

	TODAY	DECISION PLATFORM
Unit of change	Module release	Registered strategy
Time to change	Weeks to quarters	Days
Evidence	Before and after	Concurrent control, with an interval
Blast radius	Known after rollout	Counted before exposure
Reversal	Rollback deployment	Change an exposure number
Attribution	"Approval moved"	"This strategy, +N bps, [a, b]"
Learning	One-off, then lost	Compounding decision evidence

Experimentation

= the mechanism

Payment optimization

= the outcome

Decision Platform

= the capability

The goal is not faster experimentation. It is faster, evidence-backed improvement in approval, cost and payment economics.

THREE WORDS

What we mean by a strategy

DECISION POINT

A place in the payment flow where we make a choice.

Which credential to present · whether to authenticate · which processor · what to retry. Five today, extensible.

STRATEGY

How that choice gets made, for one population.

Registered, versioned, owned, and set to expire. Deployed without an engineering release.

FLOW

A complete path through those choices.

A test version differs from today's at exactly one decision — so we know what caused the outcome.

We change one decision at a time, on purpose.

Change two and a win tells you nothing about which change produced it — or whether one produced a gain and the other quietly cost more than it. The platform refuses to build a test that differs in two places.

Precise technical definitions — decision contract, binding, divergence point, propensity — are in the deep-dive deck.

WHERE IT APPLIES

One mechanism, five optimization domains

DECISION	EXAMPLE STRATEGY	BUSINESS OUTCOME
Credential	Vault network token vs merchant-supplied credential	Approval · credential economics
Routing	Processor, acquirer, network path, least-cost vs max-approval	Approval · cost of acceptance
Retry	Retry timing, alternate processor, alternate credential	Recovered revenue
Authentication	Exemption, challenge, frictionless, step-up threshold	Approval · conversion
Rail	Card vs account-to-account vs stored value	Cost · conversion
Future	Learned joint strategy across two or more of the above	Global payment economics

We do not build five optimization platforms. We build one decision platform and plug strategies into it.

The first proof point is credential selection, because it is the cleanest decision to isolate. The largest prize is routing.

THE OBJECTION WE WILL GET

"Isn't this just configuration?"

Configuration

tells you what the system did.

Necessary, and we already have it. This is not a criticism of the platform we run.

Decision Platform

tells you whether doing it was better.

Against a concurrent control, with a confidence interval and a date.

Configuration



Experimentation



Causal evidence



Learning



Optimization

where we are today

Plenty of payment platforms are configurable. Very few can prove that a configuration was better.

Configuration is necessary and we have it. The missing capability is proving whether a configuration was better — a harder thing to build than a feature, and a more durable one.

Credential selection, end to end

"For card-on-file transactions on merchants already using the Credential Vault, presenting the merchant-supplied credential instead of the vault network token will reduce approval by at least 25 bps."

SETTING	VALUE	WHY IT IS THAT
Control	vault network token · 95%	Today's behaviour for a vault client
Variant	merchant credential · 5%	Deliberately the weaker arm. Kept small
Holdout	2%, never assigned	Measures the platform, not this test
Duration	66,900 variant enrolments	A sample size, not a calendar date
Stops early if	gap exceeds -80 bps, or any guardrail	Automatic. Not a meeting

The vault is not the product. The vault experiment is the first proof point — chosen because it is the cleanest decision to isolate, and because we already sell the vault on a claim we have never measured.

Approval is one line of five

Variant is the merchant-supplied credential. A negative approval delta means the vault won.

MEASURE	CONTROL · VAULT	VARIANT · MERCHANT	DELTA	INTERVAL
Transaction approval	87.68%	87.31%	-37 bps	[-54, -20]
Cost of acceptance	—	—	-1.4 bps	—
Fraud proxy (30d)	—	—	+2.2% rel	[+0.3, +4.1]
Dispute proxy	—	—	+1.8% rel	120d pending
Credential staleness	—	—	+34% rel	—

1.4 bps of cost buys 37 bps of approval — an exchange rate of 3.8 bps per 100.

A readout that showed only the approval line would be a sales pitch, and we would have acted on it.

The same result, three different answers

What a 37 bps difference is worth, per \$1B of qualifying volume.

+\$1.16M

a year, to the merchant

+\$11K

to us — interchange-plus

−\$129K

to us — blended pricing

LINE	MERCHANT	US · IC-PLUS	US · BLENDED
Approved volume the vault saves	+\$3.70M	—	—
Merchant gross margin @ 35%	+\$1.30M	—	—
Our net revenue @ 30 bps	—	+\$11K	+\$11K
Vault cost of acceptance @ 1.4 bps	-\$140K	passed through	-\$140K
Net	+\$1.16M	+\$11K	-\$129K

This is the class of finding a dashboard cannot produce. The vault is a clear win for the merchant and, on blended pricing, something we fund and recover from nothing. That is a pricing decision — and it is now a decision with a number attached to it.

THIS IS NOT ONE EXPERIMENT

The optimization portfolio

NOW

Deterministic strategies

- Cost-aware routing
- Approval-aware routing
- Credential selection
- Retry timing and placement
- Static merchant preference

NEXT

Composed and segmented

- Joint credential and route
- Authentication and route
- Segment-specific strategies
- Merchant-specific objectives
- Recovery orchestration

LATER

Learned strategies

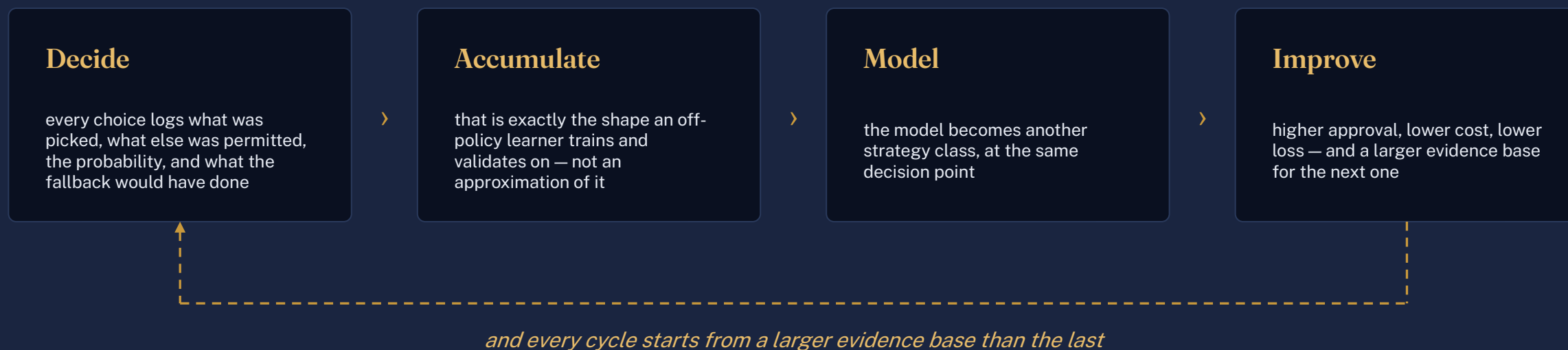
- Learned routing
- Contextual bandits
- Counterfactual policy optimization
- Autonomous optimization within guardrails
- Joint economic objective

Every item in all three columns is the same artefact, registered the same way, tested the same way and governed by the same guardrails.

THE COMPOUNDING ADVANTAGE

Every experiment pays twice

Once in the result. Again in the evidence it leaves behind — which is the only data that can train or validate a model.



Counterfactual data cannot be created backwards.

Every month we do not log propensity and the fallback action is a month of traffic that can never train or evaluate anything. Delay does not postpone the benefit by the length of the delay — it postpones it by the delay plus the three months of history a learned strategy needs before it can be evaluated at all.

Evidence is the starting point, not AI

The platform produces labelled decisions, the alternatives that were permitted, the probability assigned to each, the fallback that would have fired, the outcome, the guardrail result and the population context. Machine learning is an optimization layer over that evidence — and it has nowhere to stand without it.

Deterministic

rules and thresholds, written by a person

today

Parametric

a tuned objective with bounded parameters

near term

Learned

a trained policy, evaluated off-policy before exposure

once evidence exists

Autonomous

continuous optimization inside fixed guardrails

later, and never without them

All four run through the same decision contract, the same registration gates and the same guardrails. A model does not get a private path to production.

Without this platform a model has no safe path to production.

With it, a model is just another strategy — and the hard part was never the model. It was the evidence to train it, the gate to expose it safely, and the readout to prove it earned its place.

Three things, and only one needs a decision today

1

Propensity and fallback logging on the current engine

in flight

No behaviour change. The decision is unchanged, only the record is richer. Cannot be backfilled, and every learned strategy depends on it.

2

A verified decision-point and capability registry

calendar time

Which decision points exist, what each may choose from, and what a route can actually support — verified with processors rather than asserted.

3

A reserved permanent holdout

decision

Two percent, set aside before the first experiment runs. It cannot be created retroactively, and without it we can measure individual experiments but never the platform.

Ninety days, three outcomes

Each column is defined by what becomes true at the end of it, not by what gets worked on during it.

BY DAY 30

We can reconstruct every eligible decision and its counterfactual.

WHAT IT TAKES

- Propensity and fallback logging live on the current routing and retry engine
- Permanent 2% holdout reserved and excluded
- Decision-point and capability registry scoped and agreed
- Five ownership boundaries assigned

BY DAY 60

We can register, replay and validate a strategy without a release.

WHAT IT TAKES

- Registry built and verified with processors
- Strategy registration and flow composition, with the single-difference check
- Replay and backtest against logged traffic
- Experiment registration with mechanical gates

BY DAY 90

We have one controlled experiment running and can produce an economic readout.

WHAT IT TAKES

- Credential experiment registered and backtested
- Live exposure at 3–5%, with automatic guardrail abort
- Per-transaction decision trace, self-serve
- Readout covering approval, cost, fraud and net

Beyond 90 days: scale across merchants, add decision points, open experiment 2 on non-vault clients, and begin the first learned strategy once three months of propensity-logged history exists.

TODAY'S DECISION

Reserve a permanent 2% platform holdout

It is the only thing on this list that cannot be created retrospectively, and it is the difference between measuring individual experiments and measuring whether the platform was worth building.

LEADERSHIP ALIGNMENT REQUIRED

- | | | |
|---|------------------------|--|
| 1 | Product · Finance | Economic objective — how much cost we accept for approval, and whether it differs by pricing model |
| 2 | Product | Exploration budget — the basis points we spend learning, and the ceiling on them |
| 3 | Architecture | Decision-point registry — what becomes an experimentable decision point |
| 4 | Platform | Exposure authority — who may move traffic, and who answers when an arm aborts |
| 5 | Legal · Client Service | Exploration consent — whether a knowingly weaker arm on a named client needs contractual consent |

No funding decision today

No production rollout today

No merchant behaviour change today